

Who do we trust on climate change, and why?

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ABSTRACT

Trust in climate communicators is a critical determinant of whether the public accepts and acts upon climate change information. Yet most research to date has focused on who is trusted, with less attention to why certain messengers are deemed trustworthy. Using survey data from 6479 participants across 13 countries, this study examines (1) which sources of climate information are trusted, (2) what features make a communicator trustworthy, and (3) how these judgments differ between climate change believers and skeptics. Scientists were the most trusted sources among climate believers, but overall, the most trusted sources are informal and identity-based: “friends and family” and “people like me.” Across the sample, trust was predicted not only by demographic variables but also by specific communicator features: most notably clarity, shared values, sincerity, and being respectful of opposing views. Believers and skeptics prioritized different features, underscoring that trust is not a universal response but shaped by ideological identity. These findings reveal the layered and audience-contingent nature of trust in climate communication. By identifying the features that drive trust across different audiences, this study offers practical guidance for communicators interested in tailoring messages and messengers to more effectively engage the public on climate action.

1. Introduction

Climate change is among the most urgent and complex challenges facing humanity. As climate-related disasters intensify and threaten human, planetary, and economic health, effective public communication becomes critical to mobilizing global action. While access to climate information is greater than ever, education is not just about the content; it also depends on *who* delivers the message. Even the most scientifically accurate messages are unlikely to motivate action if the source is not trusted. Here, we focus specifically on epistemic trust as “*accepting and relying on the knowledge claims of others even when we are not in possession of complete evidence*” (p.8. Gundersen et al., 2022). Trust is particularly relevant to processing climate change information because the public must rely on experts, institutions, and peer networks to navigate complex scientific information, behavioral guidance, and policy decisions.

While prior work has been devoted to mapping trusted sources of information about climate change, less attention has been given to the specific characteristics that are associated with perceptions of trustworthiness. Recognizing the complex interdependencies between message, messenger, and audience is essential for identifying appropriate

messengers and designing messages that resonate across diverse audiences. This paper focuses on who is trusted, what makes them trustworthy, and to whom, offering a more actionable foundation for building effective climate communication strategies.

In doing so, it is useful to examine how trust judgments differ between individuals who are climate change believers and those who are climate skeptics. Prior research consistently shows that people do not evaluate information in a vacuum; instead, their existing beliefs shape how they interpret both *who* is speaking and *what* is said (Fielding et al., 2020; Kahan et al., 2012). For example, the messenger characteristics that make a communicator trustworthy to climate believers may have little sway over climate skeptics. By testing climate belief as a moderator, we can assess whether trust in sources and communication features is universal or audience-contingent, offering practical insights for tailoring effective climate messaging across ideological divides.

In sum, the present study explores three research questions.

Research Question 1: Which sources of information about climate change do people perceive as trustworthy? This question explores public trust in various types of communicators (e.g., scientists, activists, journalists, peers) to identify broad patterns in source credibility.

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Research Question 2: What characteristics of a communicator make them appear trustworthy when sharing information about climate-related issues? This question examines the specific traits that individuals associate with trustworthiness in the climate communication context.

Research Question 3: Do perceptions of trustworthy sources and communicator characteristics differ between climate change believers and skeptics? This question investigates whether climate belief moderates trust judgments, assessing whether different groups prioritize different sources and traits when evaluating trustworthiness.

We examine these questions by leveraging a unique, cross-national dataset of 6,479 participants from 13 countries, enabling the most comprehensive examination to date of how trustor and trustee factors shape trust in climate change information. Before describing the results of the study, we first situate our three questions in relevant theory and empirical research.

1.1. Trust in different sources of information about climate change

A detailed literature review of studies that specifically compared trust in various sources of climate change information is provided in Supplementary Table S1. This review highlights patterns of commonality but also shows significant variations in relative levels of trust across sources and samples. We summarize these patterns below.

1.1.1. Trust in dedicated climate experts

In this paper, we define *dedicated climate experts* as individuals and organizations that exist primarily to serve information needs around environmental and climate change issues. The original sources of information about climate change were climate scientists, who have been reporting on the impact of greenhouse gases on the atmosphere since the 1960s. The topic of climate change entered international public discourse in 1988 (IPCC, 2001) and since then, scientific publications on the topic have been doubling in quantity every 5–6 years until 2014 (Haunschild et al., 2016) with an additional 1,650,000 articles returned from “climate change” in Google Scholar since then (as of October 2025).

A meta-narrative review found that while scientists are generally well-trusted, where mistrust exists it often stems from concerns about their motives, competence, and perceived biases (Fage-Butler et al., 2022). Climate scientists have been the target of particularly high-profile attacks from public figures, with robust conspiracy theories emerging about their competence, motives, and integrity (Douglas and Sutton, 2015; Hornsey et al., 2018; Stockemer and Bordeleau, 2024; Tam and Chan, 2023). One might expect, then, that trust in climate scientists would be fragile. However, among studies that compare trust in different sources of information about climate change, items that refer to scientists or scientific organizations almost always top the list (Boon, 2016; Boulianne and Belland, 2022; Brewer & Ley, 2013; Buckley et al., 2017; Choo, 2023; Gowda et al., 1997; Hagen et al., 2016; Leiserowitz et al., 2022; Liu et al., 2023; Marchezini et al., 2022; Scott-Parker et al., 2017; Wilkins et al., 2018).

Findings from the scientific literature are enacted and distributed through the community by various environmental and climate change organizations, mostly within the government and NGO sectors. Although trust in these organizations is routinely lower than for organizations that are explicitly referred to as “scientific”, trust levels are still typically above the midpoint of scales (Brewer and Ley, 2013; Choma et al., 2016). Furthermore, several studies which required participants to rank-order trust in various sources found “environmental groups” or “environmental organizations” to be ranked in the top half (Gowda et al., 1997; Hagen et al., 2016; Leiserowitz et al., 2022; Zerva et al., 2021).

Fewer studies have explicitly asked about trust in environmental nonprofit organizations. In a sample of over 10,000 European residents, “environmental NGOs” were ranked 4th most trusted of 12 sources of

information about climate change impacts on the coastline (Buckley et al., 2017). But a sample of the Taiwanese general public nuanced this finding: environmental nonprofit organizations associated with “environmental scholars” and “local residents” were rated as highly trusted, whereas environmental nonprofit organizations associated with “celebrities” and “economists” were rated below the midpoint of the scale (Liu and Morris, 2022).

One specific form of nonprofit activity is that of climate activist groups. Although these groups play an increasingly prominent role in climate discourse – often through high-profile figures like Greta Thunberg – they are rarely included in studies that benchmark trust across multiple sources. One exception is Choo (2023), who found that North American participants rated “activist organizations” as less trusted than scientists, but more trusted than government and business leaders. Given the growing visibility of activist groups, including those that favor disruptive tactics (e.g., Just Stop Oil, Extinction Rebellion), there is a need for a more comprehensive empirical base on how they are perceived.

1.1.2. Trust in general societal leaders

Some institutions – for example government, media, business, and religious groups – do not meet our definition of dedicated climate experts but can still be important sources of information about climate change given their roles as thought-leaders on diverse societal issues. Here, we refer to these sources as *general societal leaders*.

Government figures typically perform modestly in studies that compare different sources of trust in climate change. Indeed, studies in Europe (Buckley et al., 2017), South Pacific (Scott-Parker et al., 2017), and North America (Boulianne and Belland, 2022; Choma et al., 2016; Choo, 2023; Gowda et al., 1997; Leiserowitz et al., 2022; Wilkins et al., 2018) ranked politicians, government officials, and government agencies in the bottom half of a list of most-trusted sources of information about climate change. Among Chinese farmers, government entities were rated highly (3.91 on a 5-point scale) although this was marginally lower than “scientists” and “media” (Liu et al., 2023).

Top news outlets in the U.S. have increased their coverage of climate change by 300 % from 2011 to 2023 (Bolstad and Victor, 2024). Social media postings about climate change are also on the rise, where tweets and retweets discussing weather events in relation to climate change increased from 1,676 in 2008 to 436,691 in 2017 (Berglez and Al-Saqaf, 2021). When described in ways that imply scientific integrity, trust in these forms of media is high. For example, “science television” was ranked highest of 8 sources among Milwaukee residents (Brewer and Ley, 2013) and “televised weather reports” was ranked 2nd of 9 sources in a sample of >3000 Europeans (Hagen et al., 2016). However, social media (Hay et al., 2019; Mase et al., 2015; Prokopy et al., 2015), radio (Hay et al., 2019; Zerva et al., 2021), mainstream TV channels (Leiserowitz et al., 2022; Maibach et al., 2011; Wilkins et al., 2018) and newspapers (Brewer and Ley, 2013) frequently come last or near last in lists of most trusted sources.

Other influence agents have fared even worse. For example, “industry”, “business organizations” or “industrial organizations” have been ranked last in Europe (Buckley et al., 2017) and North America (Choo, 2023; Choma et al., 2016; Leiserowitz et al., 2022). Religious leaders and religious groups have been ranked last in the South Pacific (Scott-Parker et al., 2017), Europe (Hagen et al., 2016) and the U.S. (Wilkins et al., 2018).

1.1.3. Trust in similar others

The third major source of information about climate change is what we call *similar others*: people who have social and psychological proximity to the respondent but do not have institutional authority. In trust studies, this is most frequently operationalized as friends and family. Trust in friends and family as sources of information about climate change tends to be high, ranking in the top half of all published lists of most-trusted sources (Buckley et al., 2017; Hagen et al., 2016; Hay et al.,

2019; Wilkins et al., 2018). Indeed, in a representative sample of >1000 U.S. respondents, “family and friends” was ranked the 2nd most-trusted of 20 sources of information about global warming, even higher than “climate scientists” (Leiserowitz et al., 2022). Strikingly, “family and friends” was ranked the 3rd most trusted of 15 sources of information among U.S. agricultural advisors, far higher than the IPCC and federal agencies (Mase et al., 2015; Prokopy et al., 2015). Although it might seem surprising that people would trust friends and family above expert agencies, this pattern is consistent with findings from other domains, including health and science communication, where informal interpersonal sources are often highly trusted (Allington et al., 2021; Funk et al., 2017).

1.2. Features of trusted communicators

Above, we discussed literature on trust in different sources of information about climate change. However, understanding trust in climate communication requires attention not only to *who* is trusted, but also *why* they are trusted. While past research has often examined trust in specific sources (e.g., scientists, government officials, peers) this approach alone may miss the underlying criteria people use to judge trustworthiness. For example, two individuals might both report trusting “scientists,” but one might do so because of perceived expertise, while the other is drawn to scientists’ perceived integrity or value alignment. By examining both which sources are trusted and what features are associated with that trust, we gain a more nuanced picture of what underpins trust in climate communicators.

Trust in communication is not inherent in the message or the messenger alone but emerges from their interaction with the audience. This perspective builds on a long tradition of scholarship on source credibility and persuasion (Hovland and Weiss, 1951), which first identified trustworthiness as central to communication effectiveness. Later models, like the ability-benevolence-integrity framework (Mayer et al., 1995) unpack the components of trust, showing that audiences evaluate competence, intentions and moral orientation to form their impressions of trustworthiness. Social psychological theories extend this understanding by showing that such evaluations are filtered through identity and motivation: social identity theory (Tajfel and Turner, 1979) and motivated reasoning (Kunda, 1990) reveal how group affiliations and worldview defense shape perceptions of trustworthiness.

Integrating these perspectives, we conceptualize trust as an interaction between trustor characteristics (e.g., ideology, identity) and trustee features (e.g., ability, benevolence, integrity, delivery). For instance, climate believers may privilege ability- and data-based cues, whereas skeptics may respond more to benevolence, openness, and shared values. Our study builds on this foundation by distinguishing *who* is trusted from *why* they are trusted, integrating trustor and trustee qualities to explain variation in climate communication trust.

Studying the “who” and “why” dimensions together also allows us to disentangle surface-level labels (e.g., “a journalist” or “a friend”) from the deeper attributes that make someone compelling. This is especially important in a fragmented media environment where traditional categories of authority are less stable, and where individuals may trust non-experts (like peers or influencers) if they embody the right mix of clarity, care, and value alignment. In this sense, our approach integrates classical models of credibility with contemporary identity-based approaches, viewing perceived trustworthiness as dependent not only on evidence and expertise but also on relational and moral resonance with the audience. In short, understanding trust in climate communication requires an integrated view: not just *which sources* are trusted, but *what qualities* make any source rise to the level of trustworthiness in the eyes of *different audiences*.

Below, we discuss literature on what characteristics make a person trustworthy when discussing climate-related issues. Our review clusters around four key themes: competence; shared identity and values; communication style and accessibility; and integrity. These themes

reflect distinct but complementary pathways through which communicators earn public trust.

Competence refers to the perceived knowledge, qualifications, and evidence-based reasoning of the communicator. This is perhaps the most intuitive basis of trust, rooted in early persuasion research (Hovland and Weiss, 1951) and frequently emphasized in science communication. Studies show that audiences often rely on heuristics like “expert = trustworthy,” especially when they lack technical knowledge themselves (Hovland and Weiss, 1951; Brossard and Nisbet, 2007). For example, audiences often use heuristic cues – like a PhD or reliance on credible, peer-reviewed data – to infer credibility (Druckman and Lupia, 2000).

Shared identity and values shape trust through a different mechanism: social and cultural alignment. According to social identity theory (Fielding and Hornsey, 2016; Tajfel and Turner, 1979), individuals are motivated to adopt the norms, attitudes, and beliefs of their salient ingroups, and are more likely to trust and act on messages delivered by perceived ingroup members. In the climate domain, recipients’ political or ideological identities should shape trust in messengers and their messages because these identities are strongly linked to climate beliefs (Hornsey et al., 2016). Kahan et al.’s (2012) cultural cognition thesis emphasizes that individuals assess information in ways that are congruent with their cultural identities. In other words, a trusted communicator must not just be scientifically credentialed but also aligned with the audience’s worldview. Similarly, Fiske and Dupree (2014) show that value alignment plays a major role in perceived warmth, which is a core antecedent of trust. Research shows that people are more likely to trust communicators who they perceive as “like them,” whether in terms of values, goals, or social group membership (Kahan et al., 2012; Fiske et al., 2007; Hornsey and Lewandowsky, 2022). Thus, even a non-expert may be perceived as trustworthy if they are seen as aligned with the audience in terms of their underlying priorities, values, and identities.

Communication style and accessibility further influence perceptions of trust. A communicator who uses jargon, speaks opaquely, or dismisses opposing views may be perceived as arrogant or unapproachable, qualities that erode trust even if the content is factually sound (Wynne, 1992; Besley and Nisbet, 2013). This reflects a broader shift in science communication from a “deficit model” (where experts transmit knowledge) to a “dialogue model” that prioritizes engagement, empathy, and co-production of meaning (Leach et al., 2005). Accessible communication – explaining complex topics in plain language – is linked to increased trust and public engagement (Besley and Nisbet, 2013).

Integrity refers to whether the communicator is perceived as sincere and free from ulterior motives. These cues are conceptually distinct from benevolence in trust models, which emphasizes good will (Mayer et al., 1995). Both dimensions highlight that competence alone is insufficient for trust – communicators must also be seen as honest and well-intentioned. Passion may act as a proxy for intrinsic motivation, suggesting that the communicator cares deeply about the issue and is not simply acting out of self-interest. While passion is less discussed in the trust literature, recent research suggests that affective expression (when seen as sincere) can enhance perceptions of trustworthiness, especially among audiences who are emotionally engaged (Hart and Feldman, 2014).

1.3. Patterns of trust among climate skeptics versus climate believers

While formative work shows some general patterns of trust in differing sources of climate change information, it also illustrates substantial variability depending on the political leanings and climate change belief status of the sample. Yet, how source characteristics interact with recipient characteristics to inform trust in climate change information has received less attention.

Climate change skeptics report different levels of trust overall to the general public and should therefore be treated as a distinct group in attempts to understand trust in sources of climate change information.

For example, climate skeptics generally rank the trustworthiness of climate scientists (Saratthachandra and Haltinner, 2023) and environmental organizations (Kränge et al., 2021; Krishna, 2021) lower than the general population. Meta-analyses show that trust in science is related to increased belief in climate change (Hornsey et al., 2016; Bogert et al., 2024), and trust in climate scientists is associated with increased pro-environmental behavior (Cologna and Siegrist, 2020). Additionally, trust in scientists, environmental activists, international organizations, and the news is associated with less false beliefs about climate change, while trust in politicians, celebrities, and energy companies is associated with higher levels of false beliefs (Ejaz et al., 2024).

Skeptics may also differ from believers in terms of the features of a communicator they associate with trustworthiness. For example, skeptics may distrust scientific or governmental sources not because they doubt their competence per se, but because they perceive a mismatch in values, motives, or political alignment. Likewise, the traits that make a communicator trustworthy to believers (e.g., academic credentials) may have little sway over skeptics, who may instead prioritize traits like openness, fairness, or independence from perceived “climate agendas.” These ideas are speculative, however, given the lack of empirical attention paid to this question.

1.4. The present study

Our study breaks new ground in four ways. First, we analyze trust across a globally diverse sample spanning 13 countries, enabling broader generalizability than the studies summarized in Table S1. Second, unlike many prior studies that examine only absolute levels of trust, we move beyond description to probe the underlying features of both trustees (e.g., academic credentials, supporting data) and trustors (e.g., demographics), identifying which specific messenger traits predict trust and how strongly they do so. Third, we apply multilevel modelling to examine both within-country and between-country variation, offering a more fine-grained analysis than traditional regression approaches and enabling robust estimation of effect sizes and cross-level interactions rarely reported in previous work. Finally, we examine variation in trust patterns across both climate change believers and skeptics, providing insight into whether different audiences prioritize different cues to trustworthiness. In sum, the current study extends prior work by offering statistical evidence on the patterns, predictors, and contextual variability of trust in climate communication.

2. Methodology

2.1. Study design and participants

We analyzed data from the 2023 Edelman Trust Barometer survey “Special Report: Trust and Climate Change” (<https://www.edelman.com/trust/2023/trust-barometer/special-report-trust-climate>). Data were collected from September 20 to October 4, 2023, via a self-administered, online, 25-minute, local-language survey in 14 nations: Brazil, Canada, China, France, Germany, India, Japan, Mexico, Saudi Arabia, South Africa, South Korea, UAE, U.K., and the U.S. The total sample size was 13,986 or approximately 1000 respondents per country. National samples were weighted to be representative of the general population for age, gender, and region for that country.

Respondents were split between blocks which offered item subsets of the total survey. Because of evenly split blocking and systematic branching of the survey, not all respondents were presented with the same items. We used a subset of respondents from the main survey who

Table 1
Sample demographics.

	Minimum	Maximum	<i>M</i>	<i>SD</i>
Age (years)	18	90	44.40	16.47
Country of residence			Frequency	Percent
United States			489	7.5
Canada			501	7.7
Japan			499	7.7
South Korea			500	7.7
India			500	7.7
United Kingdom			501	7.7
Germany			500	7.7
France			501	7.7
Mexico			500	7.7
Brazil			500	7.7
UAE			500	7.7
South Africa			500	7.7
Saudi Arabia			488	7.5
Gender				
Male			3364	51.9
Female			3099	47.8
Non-binary or third gender			10	0.2
Prefer not to answer			6	0.1
Income				
1 st quartile			1511	23.3
2 nd quartile			1533	23.7
3 rd quartile			1633	25.2
4 th quartile			1509	23.3
Missing			293	4.5
Education				
Primary school/equivalent or less			68	1.0
Some secondary school or equivalent			293	4.5
Finished secondary school or equivalent			1121	17.3
Vocational school/technical college			791	12.2
Some university/higher education			817	12.6
University degree/honors degree			2347	36.2
Postgraduate degree			1024	15.8
Refused/NA			18	0.3
Political leaning				
Very left (very liberal)			224	3.5
Left (liberal)			1057	16.3
Center (moderate)			2585	39.9
Right (conservative)			1321	20.4
Very right (very conservative)			280	4.3
Don't know/not sure			818	12.6
Prefer not to answer			194	3.0
Total			6479	100.0

Note: For income, participants were offered a range of income brackets representative of their country of residence. They selected their income bracket or selected “don’t know” or “refuse to answer”. The income was then transformed to rebase the percentages to exclude those who selected “don’t know” or “refuse to answer.” From there, four quartiles were determined that are split as close to 25% each as possible, leading to an income variable ranging from 1 to 4 (low to high).

had been presented with our key items of interest. One consequence of this is that the respondents from China were not analyzed (they were not asked any questions about trust in government). Excluding those who had not been presented with the focal items, the resulting survey sample consisted of 6,479 adults from 13 nations. Demographics of the final sample are summarized in Table 1.

2.2. Measures

Level of trust in sources of climate change information. Participants were prompted as follows: “Below is a list of categories of people. For each one, please indicate how much you trust that group of people to tell you the truth about climate change and about what our country should be doing to help prevent its worst consequences”. Participants responded on 9-point

Likert scales, ranging from “do not trust them at all” (1) to “trust them a great deal” (9). This question was asked separately for 14 sources: “a person like yourself”, “my friends and family”, “scientists and climate experts”, “national environmental authorities such as [prefilled for a national agency relevant to their country]”, “climate activists/advocates like Greta Thunberg”, “the heads of non-profit organisations that address climate change such as Greenpeace, the Environmental Defense Fund, and the environment congress for Asia and Pacific (ECO)”, “journalists”, “CEOs”, “social media influencers”, “government leaders”, “central/federal government leaders”, “your regional, provincial, or state government leaders”, “your local, city, or municipal government leaders”, and “religious leaders”.

Features of trustworthiness. Participants were asked to assess “When it comes to climate-related issues and information, what makes someone, in your mind, a trustworthy source?”. Respondents ticked all that applied out of the following seven options: “they have academic credentials like a PhD in a scientific field”, “they share my goals, priorities, and values”, “they back up everything they say with data and research”, “they speak in a way that is easy for me to understand”, “they are not in a position to profit from what they are saying or asking me to do”, “they speak with passion about the subject”, and “they are not dismissive of people who think differently than they do”. These variables were binary, coded so that a value of 0 indicates the feature was not identified by respondents as relevant to trustworthiness, while a value of 1 indicates it was identified as relevant.

Belief in climate change. Participants were asked “Which of the following best describes your beliefs about climate change?” and were asked to select one of the following four options: “I have always believed that climate change poses a serious and imminent threat to the planet. There has never been a time in my life when I did not feel this way”, “Over time I have come to believe that climate change poses a serious and imminent threat to the planet. But, there was a time in my life when I did not feel this way”, “I do not believe that climate change poses a serious and imminent threat to the planet”, or “Don’t know / Not sure”. Given the difficulty in interpreting the “Don’t know / Not sure” responses, we treated them as missing data. We classified skeptics as those who selected: “I do not believe that climate change poses a serious and imminent threat to the planet” ($n = 509$). Other respondents were categorized as believers ($n = 5,771$).

3. Analyses and results

3.1. Trusted sources

Fig. 1 illustrates variability in mean levels of trust between different sources separately for climate believers and climate skeptics. We summarized mean scores for all 14 sources included in the Edelman survey

and color-coded the graph to reflect the putative higher-order themes that unite the individual sources (i.e., dedicated climate experts, similar others, general societal leaders). Before doing so, principal components analysis was conducted to assess if the sources of climate change information fell naturally into these three categories. As detailed in Supplementary Table S2, this analysis did indeed reflect the three categories described in the Introduction: “dedicated climate experts” (head of climate change non-profit, scientists/climate experts, climate activists/advocates, national environmental agencies), “general societal leaders” (government leaders, regional/provincial/state government leaders, central/federal government leaders, local/city/municipal government leaders, CEOs, social media influencers, religious leaders, journalists), and “similar others” (friends/family, someone like me).

As can be seen in Fig. 1a, the sample of climate believers was relatively trusting of all sources: even the least-trusted source (social media influencers) was rated only slightly below the midpoint of the scale ($M = 4.28$; all means and standard deviations are detailed in Supplementary Table S3). Consistent with most previous research, “scientists and climate experts” was rated the most-trusted source. Interestingly, the next most trusted sources were similar others (“friends/family”, “people like me”) which were rated as more trusted than climate change non-profits, environmental authorities, and climate activists. All eight of the least-trusted sources were general societal leaders. In contrast to climate believers, only two of the sources were rated above the midpoint among climate skeptics. Both were similar others (“friends/family”, “people like me”). As detailed in Supplementary Table S4, ANOVAs revealed that skeptics trusted all sources of climate change information less than did believers ($ps < 0.001$). Effect sizes ranged from small to large, depending on the source. The largest effect was observed for trust in the aggregated “dedicated climate change experts” ($\eta^2 = 0.115$), while the smallest effects were for “friends and family” and “someone like me” ($\eta^2 \leq 0.003$), implying trust is most divergent between climate-specific sources and least for personally relevant ones.

Observing the post hoc pairwise comparisons within groups (see Supplementary Table S5), believers consistently rated scientists/climate experts as significantly more trustworthy than all other sources (all $ps < 0.001$). In contrast, skeptics rated “someone like me” and friends/family significantly more trustworthy than scientists. Scientists were, however, trusted significantly more than government leaders, social media influencers, and religious figures, as well as journalists, activists, and environmental authorities across both believers and skeptics (all $ps < 0.001$).



Fig. 1a. Descriptives of source trust for believers

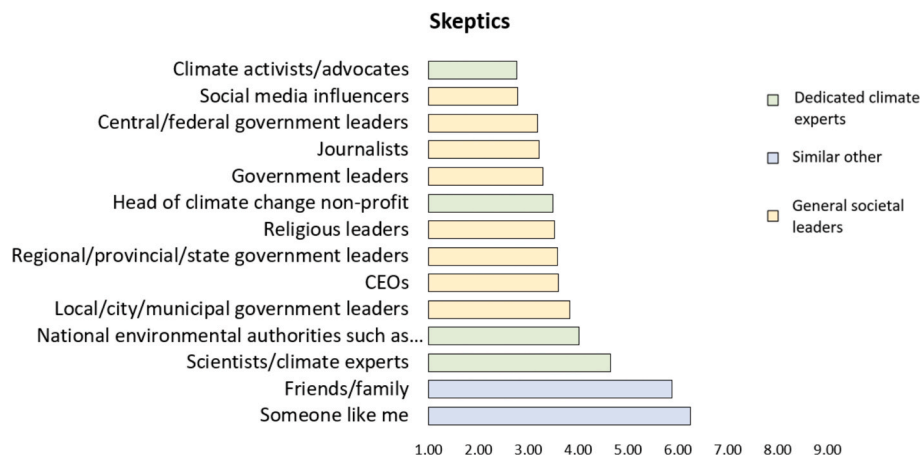


Fig. 1b. Descriptives of source trust for skeptics

3.2. Features of trustworthiness

Fig. 2 illustrates variability between frequencies of selected features important for trustworthiness. “Supporting data/research” was the most highly nominated feature of trustworthiness among both believers and skeptics. However, considerable discrepancies emerged between skeptics and believers in the frequency with which they selected the messenger characteristics as important for trustworthiness. Chi-square tests of independence revealed that frequencies were significantly different between skeptics and believers for all features of trustworthiness, except for “don’t directly profit”, which was non-significant (for summary of all Chi-square results, see Supplementary Table S6). Compared to skeptics, believers placed greater value on supporting data, ease of understanding, passionate speaking, shared goals/values, and academic credentials. In contrast, skeptics selected not dismissing opposition more frequently than believers.

3.3. Multilevel modelling predictors of levels of trust

Multilevel modelling was conducted to examine the effects of trustor and preferred trustee features on trust in climate change experts, general societal leaders, and similar others. Three separate analyses were conducted with each source group as the dependent variable: trust in dedicated climate experts (averaged across four sources: $\alpha = 0.85$), trust in general societal leaders (averaged across eight sources: $\alpha = 0.93$), and trust in similar others (averaged across two sources: $\alpha = 0.68$). Results for each model are summarized in Table 2.

The top half of Table 2 summarizes results for characteristics of the trustor: respondents’ age, gender, education, income, political leaning, and climate change belief (i.e., skeptic versus believer). These results inform us about whether trust in different sources differ as a function of respondents’ demographic backgrounds and belief in climate change. The bottom half of Table 2 summarizes results for characteristics of the trustee: the seven features of trustworthiness. These variables were

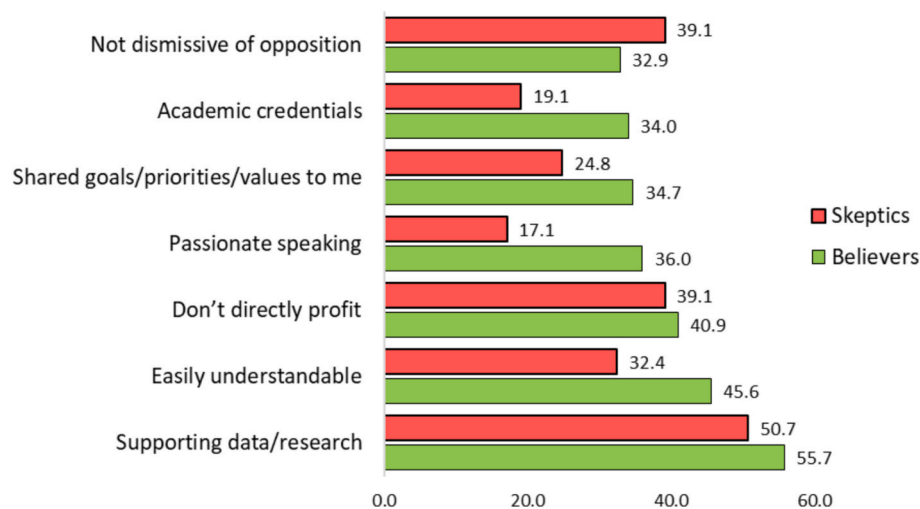


Fig. 2. Frequency of selected features of trustworthiness, expressed as a percentage of the number of skeptics and believers in the sample. Note: Participants could click multiple options and so the percentages do not add up to 100.

Table 2
Standardized coefficients of predictors in multilevel model.

Predictor	Dedicated climate experts			General societal leaders			Similar others		
	β	95%CIs	<i>p</i>	β	95%CIs	<i>p</i>	β	95%CIs	<i>p</i>
Respondent demographics									
Age	-0.03	-0.05, -0.01	.012	-0.10	-0.12, -0.08	<.001	0.02	-0.01, 0.04	.206
Gender	0.07	0.02, 0.12	.003	-0.00	-0.05, 0.05	.989	0.01	-0.04, 0.07	.607
Education	0.02	-0.00, 0.05	.074	0.03	0.01, 0.06	.009	0.02	-0.01, 0.05	.153
Political leaning	-0.08	-0.11, -0.06	<.001	0.03	0.01, 0.06	.009	0.03	-0.00, 0.05	.054
Income	0.04	0.01, 0.06	.002	0.06	0.03, 0.08	<.001	0.06	0.03, 0.09	<.001
Skeptical	-1.03	-1.12, -0.93	<.001	-0.59	-0.68, -0.49	<.001	-0.09	-0.19, 0.01	.088
Features of communication relevant to trust									
Academic credentials	0.19	0.14, 0.24	<.001	0.01	-0.04, 0.06	.675	0.01	-0.04, 0.07	.641
Shared goals / priorities / values	0.15	0.07, 0.23	<.001	0.10	0.05, 0.15	<.001	0.24	0.19, 0.30	<.001
Supporting data / research	0.18	0.13, 0.23	<.001	-0.10	-0.15, -0.05	<.001	0.07	0.02, 0.13	.011
Easily understandable	0.13	0.08, 0.18	<.001	0.09	0.04, 0.14	<.001	0.10	0.04, 0.15	<.001
Don't directly profit from recommendations	0.03	-0.03, 0.08	.328	-0.13	-0.18, -0.08	<.001	0.07	0.01, 0.13	.014
Passionate speaking	0.29	0.24, 0.34	<.001	0.17	0.11, 0.22	<.001	0.23	0.17, 0.28	<.001
Not dismissive of opposition	-0.09	-0.14, -0.03	.001	-0.06	-0.11, -0.00	.033	-0.04	-0.10, 0.02	.159

Note: Gender is coded as 0 = man, 1 = woman. Political leaning is coded as 1 = very left / liberal, 5 = very right / conservative. The 7 variables listed under the heading "Features of communication relevant to trust" were coded such that 0 = not selected as relevant, 1 = selected as relevant.

dichotomous, coded such that 0 indicates this feature was not selected as relevant for trustworthiness by respondents and 1 indicates this feature was selected as relevant. These results inform us about whether trust in different sources is associated with respondents' indications about what qualities of a messenger are important to them for signaling trustworthiness.

In these analyses, random intercepts were fitted for nationalities, thus education, political leaning, and age were subsequently cluster mean-centered by nationality. Analyses were performed using R Statistical Software (v4.4.0; R Core Team, 2024), with the lme4 package used for linear modelling effects (Bates et al., 2015) and lmerTest to provide *p*-values for fixed effects in linear mixed models (Kuznetsova et al., 2017). Following recommendations by Barr et al. (2013), random slopes were initially fitted for each focal association to account for potential cross-national variability. Slopes were retained unless their inclusion led to convergence failure. Each final random-slope model was then compared to its corresponding random-intercepts-only model using a likelihood-ratio test (ANOVA). For the models predicting trust in similar others and trust in general societal leaders, random slopes did not improve model fit: the former failed to converge, while the latter did not differ significantly from the random-intercepts-only model, $\chi^2(5) = 10.85$, $p = 0.054$. For the model predicting trust in dedicated climate change experts, inclusion of random slopes significantly improved model fit, $\chi^2(2) = 17.95$, $p < 0.001$, indicating meaningful cross-national heterogeneity for how one predictor (shared values) relates to trust in climate experts. Table 2 summarizes standardized effects: for full summary statistics including unstandardized effects, see Supplementary Tables S7-9.

3.3.1. Trust in dedicated climate experts

Five of the demographic variables independently predicted the extent to which respondents reported trust in climate change experts. Unsurprisingly, trust in dedicated climate experts was significantly lower among respondents who self-identified as climate change skeptics, which was by far the largest predictor. However, there were also reliable tendencies for trust in dedicated climate experts to be higher among younger respondents, women, and respondents with higher incomes and more left-wing political leanings. Interestingly, the extent to which dedicated climate experts were considered trustworthy was not associated with respondents' own levels of education.

Respondents high in trust in dedicated climate experts were more likely to nominate the following communicator features as relevant to whether they trusted a messenger: academic credentials, using supportive data, shared values, and providing easily understandable information. They were less likely to trust this group if they found it

important for a communicator to not be dismissive of opposition. The most notable effects were for "passionate speaking", which was comfortably the largest predictor, and "don't directly profit from recommendations", which was the only speaker characteristic to not predict the extent to which dedicated climate experts were considered trustworthy. Visual inspection of the country-specific slopes (Supplementary Fig. S11) indicates that Japan shows the steepest slope for shared values, suggesting that this predictor has the strongest effect on trust in climate change experts in Japan compared with other countries.

3.3.2. Trust in general societal leaders

Trust in general societal leaders was significantly lower among individuals who self-identified as skeptics, and again this was easily the biggest demographic predictor. Younger people were more likely to trust general societal leaders, as were those with higher income, higher education levels, and more conservative political leanings.

Trust in general societal leaders was significantly higher when respondents perceived that the following features were important for trustworthiness: shared values, providing easily understandable information, and passionate speaking. Trust in general societal leaders was significantly lower when respondents indicated that it was important for sources to support claims with data and research, for the source to not directly profit from recommendations, and not being dismissive of the opposition. No significant relationship was found with the tendency to value academic credentials.

3.3.3. Trust in similar others

Trust in similar others was greater among respondents with higher incomes. Interestingly, this was the only demographic variable to emerge as a significant predictor. In terms of the speaker characteristics, the biggest predictors were whether respondents identified shared values and passionate speaking as important to trustworthiness. Trust in similar others was also significantly higher among those who valued supporting data and research, easily understandable information, and the source not directly profiting from recommendations. No significant relationships were found for valuing academic credentials and not being dismissive of opposing views.

4. Discussion

What makes someone a trustworthy source of information about climate change? In an era of contested expertise and growing polarization around climate science, this question has become central to both academic and applied work in climate communication. While much

attention has focused on *what* messages are persuasive, an emerging literature highlights the critical role of *who* is delivering the message and *how* the message is delivered. The current study provides a picture of the patterns of trust in climate communication, leveraging a dataset of 6,479 participants from 13 countries to perform an inferential, cross-national analysis of why particular communicators are trusted, identifying which specific features of trustworthiness predict trust among different audiences. By formally comparing climate believers and skeptics, we also reveal distinct patterns of trust formation that have not been empirically demonstrated at this scale. Together, these contributions extend earlier work by moving beyond what sources are trusted to explain how and for whom trust exists.

A narrative about a “crisis in trust” in science and scientists has received considerable attention in recent years (Achterberg et al., 2017; Millstone and van Zwanenberg, 2000). Indeed, analysis of U.S. data shows that trust in science has declined over the last two decades, particularly among conservatives (Lupia et al., 2024; Milkoreit and Smith, 2024). However, this decline has come from a high base: most recent evidence suggests that science is among the most-trusted institutions globally (Cologna et al., 2025). In line with this broader pattern, and consistent with several studies in the climate change literature, climate scientists were the most trusted sources of information in our sample. Trust among climate skeptics was considerably lower, but even among this sub-sample scientists were the 3rd most-trusted of 14 sources of information about climate change. National environmental authorities also received relatively high ratings, but for other dedicated climate voices the pattern was more mixed. For example, climate change nonprofits and climate activists were among the more trusted sources of climate change information among believers but were rated poorly by skeptics.

General societal leaders (i.e., government figures, CEOs, social media influencers, religious figures, and journalists) were the least-trusted sources of information for climate believers, although among skeptics they fared better. Interestingly, among skeptics, trust in government figures was highest the more local the government figure. There were other discrepancies in the demographic markers of people who trusted dedicated climate experts compared to those who trusted general societal leaders. For example, dedicated climate experts were more likely to be trusted by liberal-leaning respondents, whereas general societal leaders were more likely to be trusted by conservatives.

One of the most notable findings was the high level of trust placed in “friends and family” and “people like me” as sources of climate-related information. Furthermore, trust in these sources was high among a broad cross-swaathe of the community, equally high regardless of respondents’ gender, age, education, and political leaning. This pattern aligns with broader research showing that in contexts of low institutional trust or ideological contestation, individuals often turn to informal, socially proximate sources for guidance (Kahan et al., 2012). For skeptics in particular, trust appears to be grounded less in institutional expertise and more in relational trust.

Some differences in trust between skeptics and believers may reflect motivated reasoning (Kunda, 1990), as they seek to defend beliefs that are central to their social identity. From this perspective, skeptics’ emphasis on non-dismissiveness may not only signal a preference for open dialogue but also serve as a means of protecting their worldview and group identity from perceived challenges. Alternatively, climate change skeptics may simply encounter more dismissive communications, given that their views run counter to the mainstream scientific consensus, making them more sensitive to cues of condescension. Regarding skeptics’ different preferences for sources, social identity theory (Tajfel and Turner, 1986) suggests that people derive meaning and self-esteem from their group memberships, and skeptics’ distrust of scientists or activists may thus reflect loyalty to ingroup norms rather than generalized distrust. Believers, by contrast, may value dedicated climate experts in part because they are perceived as aligned with their own ingroup identity.

These identity-based differences also help explain one of the more novel aspects of the current paper, which was being able to track the features of a messenger that were associated with trustworthiness. Here, there was one point of consensus: more than half of respondents wanted messengers to supply supporting data and research. However, only 19 % of skeptics indicated academic credentials to be an important marker of trustworthiness (compared to 34 % of believers) suggesting that skeptics were more alive to forms of data that lay outside academic consensus. To the extent that respondents valued supporting data, they were more trusting of dedicated specialists and less trusting of general societal leaders.

Other highly valued features of trustworthy messengers (in decreasing order) were that they were easily understandable, did not directly profit, were not dismissive of opposition, had shared values, and spoke passionately. In some cases, preference for different features mapped on to trustworthiness of different sources. For instance, individuals who valued supporting data and research were *less* likely to trust general societal leaders but *more* likely to trust dedicated climate change experts. Similarly, valuing academic credentials significantly predicted greater trust in dedicated climate experts, but not in general societal leaders. This finding may reflect a congruency heuristic of assessing expertise by alignment between qualifications and role (e.g. academic credentials with academic roles).

4.1. Theoretical implications

4.1.1. Moving beyond categorical source labels

Broadly, our findings support the case for moving beyond source labels in climate communication research to explore the deeper psychological attributes that underpin trust. Our findings highlight that trust in sources is not only a function of categorical identity (e.g., scientist vs government leader), but is shaped by features such as clarity, value alignment, sincerity, and independence. This supports multi-dimensional models of trust (e.g., Mayer et al., 1995), which emphasize ability, benevolence, and integrity as distinct routes to earning trust. It also underscores the need for future trust research to consider these underlying perceptions, rather than treating source labels as homogeneous entities. By modelling how these attributes predict trust, we demonstrate that trust dimensions are not merely conceptual but observable and quantifiable across contexts.

4.1.2. Identity and ideology as filters of trust

Our results are consistent with the cultural cognition thesis (Braman and Kahan, 2006; Kahan et al., 2012), showing that beliefs about climate change shape not just trust in message content, but in communicators themselves. The stark contrast between believers and skeptics in who they trust and in what makes a communicator trustworthy suggests that trust is filtered through identity and ideology. Importantly, this also supports social identity-based approaches (Fielding and Hornsey, 2016; Tajfel and Turner, 1979), which argue that acceptance of climate information is closely tied to whether the communicator is perceived as part of the ingroup. In the climate domain, this implies that recipients’ political or ideological identities shape trust in messengers and their messages.

4.1.3. Reframing the messenger-audience relationship as relational

The strong faith shown in similar others invites a reframing of the messenger-audience relationship in climate science as not just informational but relational. That is, trust emerges less from the possession of knowledge and more from the perceived intent, clarity, and cultural closeness of the messenger. This supports broader moves within science communication toward dialogical and co-produced models of engagement (Wynne, 1992; Leach et al., 2005), and highlights the value of integrating psychological, sociological, and communicative models of trust.

4.2. Practical implications

4.2.1. Mobilizing trusted lay networks

Most notably, we show that trust in climate information is not limited to institutional experts. “Friends and family” and “people like me” were among the most trusted sources overall, and the most trusted among climate skeptics. Rather than dismissing these sources as uninformed, communicators and policymakers might leverage this finding to inform communication strategies. Specifically, communicators may consider how trusted lay networks can be mobilized as bridges to more evidence-based perspectives, for example, by engaging and equipping trusted lay networks such as peers and community leaders as key climate messengers, known as the “block leader” approach (Hopper and Nielsen, 1991). This may include training programs for community advocates, partnerships with local organizations, and toolkits for peer-to-peer engagement that emphasize accessible, evidence-based communication. Peer-to-peer models may prove especially effective in audiences that are skeptical of formal authority.

4.2.2. Supporting peer-to-peer and identity-based communication

Hornsey and Fielding (2020) argue that skepticism and inaction are often less about misunderstanding the content of climate change messaging and more about identity-driven rejection of the messenger. Similarly, Fielding et al. (2020) demonstrate that climate policy support increases when policies are endorsed by political ingroup members and aligned with ingroup values, highlighting the strategic advantage of ‘ingroup messengers’. These findings highlight the strategic potential of ‘ingroup messengers,’ whose shared identity can enhance receptivity and reduce defensiveness. This aligns with our findings, which show that skeptics are most likely to trust similar others and friends/family. On the downside, the strong faith in similar others, particularly among skeptics, raises the challenge of how to disseminate accurate climate information within networks that are high in trust but potentially low in scientific literacy. Institutions and practitioners should therefore pair empowerment with safeguards – providing continuous support, access to credible sources, and clear guidance on misinformation management. Maintaining structured mentoring and feedback mechanisms can help ensure that local communicators remain connected to scientific and policy expertise.

4.2.3. Enhancing communicator credibility and tone

Our data also suggest that communicators should be attentive to how they come across; not only as competent, but also as clear, respectful, and authentic. While traditional science communication has focused heavily on credentials and data, our results indicate that traits like not being dismissive of opposition, sharing values, and speaking with passion are powerful signals of trustworthiness. The importance of being seen as non-dismissive highlights the importance of tone and humility in public communication. By contrast, current political and media discourse often lean toward dismissiveness and hostility which can inflame ideological fault lines (Brüggemann & Meyer, 2023). From our findings, climate communicators should seek non-dismissive forms of engagement to avoid fostering distrust and disconnection, for example, by cultivating two-way dialogues that model respect for disagreement and demonstrate genuine listening.

4.2.4. Tailoring messages and messengers to audience needs

Finally, communicators should recognize that trust-building is not one-size-fits-all – it depends on the dynamic interplay between delivery, source, and audience. Climate skeptics and believers differ markedly in the cues they use to evaluate trustworthiness. Efforts to engage skeptical audiences will likely be more successful when messages are delivered by people who are relatable, value-aligned, and free from perceived vested interests. For believers, emphasizing data in an easily understandable way is likely to be well-received. Our findings point to tailoring both the messenger and the message as being critical to effective climate

engagement.

4.3. Limitations and future directions

This study has several limitations that offer opportunities for future research. First, while our sample was large and cross-national, the data are correlational and self-reported. As such, we cannot draw causal conclusions about the direction of the observed relationships. Second, the survey was administered online and is not necessarily representative of all demographic groups within each country. This is a key reason we did not attempt cross-national comparisons in our analyses. Consistent with this, our multilevel models revealed that cross-national differences in the relationships between predictors and trust were generally minimal. Random slopes did not improve model fit for trust in similar others or general societal leaders, suggesting broadly consistent trust dynamics across national contexts. Indeed, “people like me,” “climate scientists,” and “friends/family” were consistently ranked among the top four trusted sources in all countries (see [Supplementary Fig. S10](#)). However, for dedicated climate experts, the inclusion of a random slope for “shared values” significantly improved model fit, indicating that the effect of this predictor on expert trust varied meaningfully across countries. Visual inspection of the country-specific slopes ([Supplementary Fig. S11](#)) suggests that this effect was strongest in Japan, where the slope for shared values was steepest. Future studies could replicate and extend these findings using nationally representative samples or through in-depth qualitative work exploring local trust dynamics. Third, our measures of trust and trustworthiness were necessarily constrained by the items included in the Edelman survey. While we captured a diverse set of traits, other important predictors (e.g., emotional tone, source familiarity, or institutional affiliation) were not measured. Future studies could broaden the set of features assessed, or explore how combinations of traits (e.g., clarity and shared values) interact to shape trust. Finally, although we explored climate belief as a moderator, future work should investigate other potential moderators of trust, such as religiosity or media use. Understanding these boundary conditions is critical to designing interventions that resonate with diverse audiences and can effectively counter misinformation.

4.4. Conclusions

In the face of accelerating climate risks, building public trust in climate information is both a scientific and societal imperative. This study reveals that trust is shaped not just by who delivers the message, but by how they deliver it and whether they are seen as relatable, sincere, and value-aligned. While scientists remain highly trusted among many, informal networks, particularly friends, family, and “people like me”, play a central role. By unpacking the features that underpin trust across different audiences, this research provides a roadmap for more effective, inclusive, and psychologically attuned climate communication.

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6. Declaration of use of AI

During the preparation of this work the authors used ChatGPT in order to streamline and reduce text. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRedit authorship contribution statement

Sarah MacInnes: Writing – review & editing, Writing – original

draft, Formal analysis, Conceptualization. **Matthew J. Hornsey:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Christian Bretter:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Samuel Pearson:** Writing – review & editing, Writing – original draft, Formal analysis, Conceptualization. **Kelly S. Fielding:** Writing – review & editing, Conceptualization. **David Bersoff:** Writing – review & editing, Resources, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gloenvcha.2025.103096>.

Data availability

The authors do not have permission to share data.

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